

Chapter 1

Teaching translation students about data in the age of generative AI

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AI translation tools are now a key part of translation education, but many educators are searching for effective ways to teach the essentials of these tools to students with no background in computer science. This chapter explains why corpora make a good entry point to learning about AI translation tools, and it explores how science communication techniques such as framing, analogies and visualization can be used to help translation educators and students come to grips with data and machine learning.

1 Introduction

Though translator education often takes place in an arts or humanities faculty, technology has been a vital part of the translation profession, and hence of translator education programmes, for at least thirty years. Over this period, translators and translation students have shown remarkable resilience as they adapt to new tools and new technology-based ways of working. However, this does not mean that teaching technologies to translation students is easy, and educating the educators can be a particular challenge (Bowker2023; Kenny2020). As the pace of new tool releases gets faster, it can be difficult for translator educators to know where to begin. This chapter proposes that teaching translation students about data is one of the key building blocks in preparing them to use translation technology effectively. In the case of translation, data frequently takes the form of texts organised into corpora. This chapter therefore begins with a brief review of corpora and corpus-based tools, noting how AI translation tools have influenced

the nature of corpora. Next, the chapter outlines why a science communication approach, rather than traditional scientific communication, could be useful in the context of translation technology education. Three science communication techniques – framing, analogies and visualisation – are combined with examples to demonstrate how science communication could be adapted to teach translation students about data-related topics in the age of AI.

2 Corpora and corpus-based translation tools

2.1 The changing characteristics of corpora in the age of AI

Over the past few decades, translators have seen the introduction and integration of a range of different tools, including concordancers (Zanettin2023), term extractors (KorkontzelosAnaniadou2022), translation memory systems (MelbyWright2023), machine translation systems (Way2020), and most recently, generative AI (GenAI) systems (Siu2024). The various tools have become increasingly sophisticated with regard to their capabilities and their underlying architecture, but a common feature for all of the tools named above is that their core functionality revolves around processing data in the form of texts. Collections of texts are usually referred to as corpora, and these can take different forms depending on the nature of the texts and the way that they are organised (McEnery2022). Translation tools that process corpora are frequently described as corpus-based tools or data-driven tools (WangEtAl2022).

As pointed out by IsabelleEtAl1993, “existing translations contain more solutions to more translation problems than any other available resource”. Therefore, one type of corpus that is used very often by translation tools is the bilingual parallel corpus (Simard2020). In this type of corpus, a collection of source texts are aligned – usually at sentence level – with their counterpart target texts. In other words, each sentence in the source text is linked to its corresponding translation in the target text. Some translation tools might also use monolingual corpora of original texts in the source and/or target language to act as a linguistic model for that language.

For students who have already learned about concordancers, term extractors or translation memory systems, the notion of a corpus is already familiar because the corpus is a very visible resource in such tools. When using these tools, the translator or translation student often has a hand in creating the corpus or may need to upload the corpus that has been provided by the client or educator. Owing to their familiarity, corpora make a good starting point for learning about neural machine translation or GenAI tools.

According to Bowker and Pearson2002,

A corpus can be described as a large collection of authentic texts that have been gathered in electronic form according to a specific set of criteria. There are four important characteristics to note here: ‘authentic’, ‘electronic’, ‘large’ and ‘specific criteria’.

While the general notion of a corpus remained relatively stable in the period before AI translation tools appeared, the introduction of these tools has ushered in some changes with regard to the features of corpora. Therefore, it is important for translation students to understand how AI tools have influenced and altered the nature of corpora.

2.1.1 Machine-readable form

The need for the corpus to be in electronic or machine-readable form has not changed. Indeed, we could say it is more important than ever since AI translation tools take on an even greater degree of corpus processing than do tools such as concordancers or translation memory systems. While concordancers and translation memory systems conduct pattern matching and then sort and display information for the tool users to interpret, AI translation tools go further by attempting to interpret the results and present fully formed translation solutions.

2.1.2 Size

Corpora are used to reveal linguistic patterns, which only become apparent when there are multiple examples of a given linguistic phenomenon. Therefore, another feature of corpora is that they are usually very large collections of text. However, our understanding of what constitutes “large” has evolved over time. The first generation of corpora created in the 1960s contained hundreds of thousands of words and were mainly consulted by linguists (McEnergy2022). These linguists used corpus analysis tools (e.g. concordancers) to help them sort and display the text data, but the linguists were still responsible for interpreting it. Today, the corpora used to power AI tools contain hundreds of billions of words (Hughes2023). This is in large part because these AI tools do not understand text in the way that people do, and they need a much larger number of examples in order to predict patterns with confidence. However, this need for extremely large corpora is influencing other characteristics of corpora, such as the texts selected for inclusion (see 2.1.3) and sometimes even the authenticity of the texts (see 2.1.4).

The availability of machine-readable texts can differ dramatically from one language to another, creating disparities with regard to an AI tools’ performance

in different languages. A high-resource situation occurs when it is relatively straightforward to gather a large number of high-quality resources for a given language or language pair. For instance, both English and French are widely used languages, and there is a lot of translation activity between these two languages. As a result, it is not too difficult to compile monolingual and bilingual parallel corpora for these languages, and they are thus referred to as high-resource languages. In contrast, a low-resource situation can occur when languages (or language varieties) are less widely used, such as some of the Indigenous languages of the Americas or the variety of French used in Canada. For languages or language varieties of limited diffusion, it can be more challenging to build a large corpus. Moreover, even if two languages have a large number of speakers (e.g. Russian and Hindi), there may not be a lot of translation activity between them, making it hard to create bilingual parallel corpora for this language pair. Therefore, languages, language varieties or language pairs for which there are few corpora available are described as being low resource.

2.1.3 Specific criteria

As emphasised by **McEnery2022**, in order to be most useful, a corpus cannot consist of texts that have been gathered at random or in a purely opportunistic way. Rather, the texts in a corpus are selected because they correspond to specific criteria and are representative of a larger set of texts with those characteristics. One clear criterion in the context of translation is that the texts should be of high quality. Beyond this, there are many different options for designing a corpus depending on its intended purpose, but the key point here is that the choice of which texts to include is motivated. For example, in the context of a corpus to be used for translation, it could be important to select texts that are on a given topic, of a certain text type or register, or from a particular time period.

Evidence of the importance of corpus design can be seen in the way that translators construct translation memory databases, such as by creating different databases (or adding relevant metadata) for different domains or for different clients (e.g. to respect their preferred terminology or house style). In this way, they can restrict a search to texts that have specific features (**MelbyWright2023**). Likewise, neural machine translation tools are known to achieve better quality when the corpus is adapted for a specific domain (**ChuWang2018**). However, as noted in section 2.1.2, AI translation tools need to have an enormous number of texts in the corpus. As a result, it can be challenging to achieve the necessary size while trying to be selective about the content. The low-resource situation

described in section 2.1.2 can be further complicated when it comes to finding certain text types or texts on very specialised topics in less widely used or translated languages and language varieties. A consequence of not having a large enough high-quality corpus is that the AI tool does not have enough good examples to draw on, and so the tool's performance may be poorer in low-resource situations. If lower quality texts are included in the corpus in order to increase the size, then the tool may generate low-quality translations. The implications of insufficient quantity and quality of texts in training corpora used by AI translation tools is discussed in more detail in sections 4.2. and 4.3.

2.1.4 Authenticity

Finally, the need to have authentic texts in the corpus used to be sacrosanct. For instance, **McEnery2022** describes a corpus as “a large body of linguistic evidence composed of attested language use” (??) and “a collection of naturally occurring language data” (??). In the case of translation, there is a desire to have high-quality data, which means using texts that have been translated by language professionals. As observed by **Kenny2011**, the reason that the developers of translation tools use corpora of human translations to train their systems is because such corpora are assumed to contain good answers to translation problems; and they are assumed to contain good answers precisely because they contain translations performed by human beings.

One consequence of the need for bigger and bigger bilingual parallel corpora for training AI translation tools has been that such authentic high-quality human translated texts have become increasingly valuable commodities. This in turn has raised many ethical questions about ownership of translation data and permission to use it, prompting an explicit need to discuss these issues with translation students. **Moorkens2022** contains a detailed examination of such ethical issues, along with suggestions for how these can be integrated into translator education (see 4.1).

Another response to the potential shortage of text needed for constructing very large corpora for use with AI translation tools has been to set aside the long-established tradition of using authentic data and to explore the use of synthetic data. In the context of translation, synthetic data is created by using a machine translation tool to translate additional texts, and then adding these machine-translated texts to the corpus (**SennrichEtAl2016**). The quality of synthetic data can vary, and while the need for additional data tends to be for low-resource languages, these are the languages for which machine translation already underperforms. Therefore, using a low-performance tool to generate more translations

in that same language is likely to result in poor quality text. If this poor quality text is then used to further train the tool, there is a risk that using synthetic data could perpetuate a cycle of mediocrity.

This section has elaborated why understanding corpora is important for translation students, and has outlined some of the ways that corpora are changing in the age of AI. The following section moves beyond what students need to know about translation data (i.e., corpora) to consider how educators can communicate this information effectively.

3 From scientific communication to science communication

3.1 Scientific communication

Scientific communication (sometimes called internal science communication) is a type of expert-to-expert communication (Hanauska2019). It occurs when one subject field specialist addresses another, with both parties having a deep knowledge of the complex matter in question. Academics are typically very comfortable with this type of communication since they spend much of their time discussing their research with their peers. Even as part of their teaching, they are normally communicating information directly from their own field, helping their students to acquire the necessary expertise to become specialists in their own right. However, the arrival of AI tools – more than any other technology – has changed that situation for many translation educators.

As noted in the introduction, technology is not new either to the translation profession or to translator education programmes. Yet as pointed out by Kenny2020, when it comes to translation technology, identifying what to teach has often been easier than working out how to teach it. As the field of translation became more technologised, some educators admitted to feeling challenged by the demands of keeping up with technology (e.g. Kenny2007; MarshmanBowker2012), but this responsibility lay primarily with those educators who had chosen to specialise in technology. Technology-related teaching was often restricted to a core course on translation technologies, where a large focus was on learning how to use the tools, which offered an array of sophisticated features (Bowker2023).

AI translation tools are different in several respects. Firstly, the user interface is comparatively simple – often requiring the user to do little more than select the source and target languages and paste or type a text. In some cases, the

AI tool may even be called automatically by another tool. Therefore, the “how-to” aspect of translation technology teaching is no longer a principal focus. A more important difference, however, is that the technology behind the interface is far from simple. As **Kenny2018** points out, older generations of technology were relatively transparent and comprehensible in their inner workings, while AI translation tools are more opaque in that the people using them – and even the people developing them – cannot always understand how the tool arrives at its proposals or output. In the field of AI more broadly, this opacity has led to a push for more explainable AI (XAI) (**Ridley2024**), while machine translation researchers are also beginning to work on explainable aspects of this technology (e.g. **LankfordEtAl2023**). In the meantime, AI translation tools are now starting to appear in courses across the translation education curriculum rather than solely in core technology courses, and translator educators are seeking ways to help themselves and their students understand how these tools can affect translation processes and products, and how to use the tools responsibly.

In these circumstances, scientific communication is not feasible since translation educators and students are not computer scientists. The technology behind neural machine translation and GenAI tools is highly sophisticated. The tools may appear to be simple because the user interface is simple, but behind the scenes, there are very complex artificial neural networks, word embeddings, vectors, transformers, and more. It is not necessary for most translation educators or students to understand all the details of how an artificial neural network operates in order to use AI-based translation tools.¹ However, understanding the role of data (i.e., corpora) in a data-driven AI translation tool is valuable and can help translation students to be more informed and responsible users of this type of technology. At the very least, understanding the role of data is a useful first step, and the approach proposed in section 4 is intended as way to introduce data on programmes aimed at educating students who will work primarily as language professionals rather than as technology specialists. But if scientific communication is not a feasible approach, can science communication help?

¹This is currently true for many translation-oriented jobs and translation education programmes. However, as **Briva-Iglesias** and **O’Brien2022** point out, new jobs are emerging that straddle the boundary between translator and technologist and that require a deeper grounding in technologies. To address this changing market, new education programmes are also emerging that place more emphasis on technological skills, such as the MSc in Translation Technology at Dublin City University in Ireland, and the MSc in Multilingual Digital Communication at McGill University in Canada. In these more technology-oriented programmes, a deeper understanding of the underlying architectures is both required of the educators and offered to the students, and data literacy training may also be deeper and more technical, such as the approach described by **HackenbuchnerKrüger2023** as part of the DataLit^{MT} project.

3.2 Science communication

When translation students hear people talking about AI-based translation tools using terms such as transformers or word embeddings, they may feel intimidated and lack confidence in their ability to use these tools effectively. Therefore, when introducing these tools, translation educators may find it helpful to draw on science communication techniques. As pointed out by **BurnsEtAl2003**, science communication is tricky to define precisely, and there are myriad competing and overlapping descriptions of this concept in the literature. However, there is general agreement that the essential goal is to make expert knowledge accessible to non-experts. This is why this activity is sometimes described more specifically as external science communication (**Hanauska2019**). In the case of translation students, while they are developing expertise in translation, they are not experts in AI or machine learning.

Academics in all disciplines are increasingly encouraged to engage in science communication to share the results of their research with a wider audience, which could include policy makers, participants in research studies, funding agencies and even the general public (e.g. National Academies of Sciences, Engineering, and **Medicine2017**). Yet while science communication is strongly encouraged, many academics receive little training in this area, even though it is a complex and skilled task (**Borowiec2023**). Fortunately, tips, guides and recommendations are emerging to help fill this gap (e.g. **Borowiec2023**; **CookeEtAl2017**; **Henville2020**). From these, we can glean several suggestions that can be integrated into teaching to help explain data-related concepts to translation students as a foundation for learning about AI translation tools.

- **Frame your message:** Frame your message in terms that are accessible, relatable and meaningful for your audience. Framing is not about marketing your point of view but about finding a way to actively engage the audience with an issue by showing them why they should care about it.
- **Make the abstract concrete and connect unfamiliar concepts to familiar ones:** Use metaphors or analogies that connect at the level of underlying commonalities to help the audience understand an unfamiliar or abstract concept in relation to a familiar and concrete one.
- **Visualise the content:** Use visual aids (e.g. images, concept maps, graphs) to support your message. Making the information available in different modes or formats can help to crystallise the ideas in the mind of your audience.

The following section considers how these three science communication techniques could be implemented to help translation educators teach concepts related to data.

4 Applying science communication techniques to teach translation students about data

4.1 Framing the message about data

It was noted in section 2 that corpora can be a good entry point for introducing data-driven AI tools to translation students because they have likely been exposed to corpora when using other tools (e.g. concordancers, term extractors, translation memory systems). Corpora are less immediately visible in the more automated AI translation tools, but this makes it all the more important to actively discuss them. As **Kenny2011** emphasises, corpus-based tools cannot function without corpora. If no corpus is present, the tool itself has nothing to offer as a translation aid. This is a very important point to share with translation students because it demonstrates the essential contribution of translators. Of course the tool developers play a key role in both programming the tools and determining how corpus data can be processed effectively, but the data itself – the translated text – is indispensable, and this fact is often glossed over when the tools are described or promoted. Yet as part of translator education, it is important to make students aware of the value of corpora because this helps them to understand their own worth as language professionals in addition to understanding how corpus-based or data-driven tools work, and where the potential pitfalls might lie.

As mentioned in section 2.1.4, translation corpora have become valuable resources, leading to some questionable practices. Framing the discussion of AI translation tools as part of a broader issue of professional practice, ethics, and digital citizenship, rather than simply as an instrumental means of making their job easier, can help translation students to recognise not only the value of their work, but also the fact that technology does not exist in a vacuum. Discussions about the role of data in data-driven tools can encourage translation students to reflect on how AI translation technology can impact tool users and end users of translated products in various ways. It also empowers language professionals to be able to participate in broader conversations about AI technology development and use in other areas of life.

4.2 Connecting unfamiliar concepts to familiar ones

The concept of data is relatively abstract. For instance, some popular dictionaries define data in the following ways:

- information in an electronic form that can be stored and processed by a computer (Cambridge Dictionary)
- information in digital form that can be transmitted or processed (Merriam-Webster Dictionary)

Of course we can make this abstract concept more concrete in the context of translation by explaining that the data in question is text, and usually a series of source texts aligned with their corresponding translations (i.e., a bilingual parallel corpus) (see 2.1). But can we find an even clearer analogy to help students understand the nature of the relationship between the data (corpus) and AI-based translation tools?

In discussions about AI more broadly, data is frequently compared to oil or water. However, as pointed out by **Kenny2011**, this is problematic because it suggests that data is a naturally occurring resource that can be exploited (e.g. through harvesting or mining), whereas in reality, translation data has been created by skilled translators who deserve to be recognised for their contribution. As part of a discussion about why the metaphor of natural resources is not appropriate for describing data, translation educators and students can explore where translation data comes from, and what might happen to it after a translator delivers it to a client. For such conversations, translation educators can be inspired by **Moorkens2022**, who undertakes an in-depth examination of the concepts of translation data ownership, permission to use translation data, and related questions concerning the ethical use of AI translation tools.

While the metaphor of translation data as a naturally occurring resource is clearly problematic, an analogy such as fuel for a vehicle could be useful for helping translation students to understand the relationship between the translated texts and the AI translation tool. Most car drivers do not have a detailed understanding of how their car's engine works. However, they do understand that it is necessary to keep fuel in the tank, and they know that it is important to select the right type of fuel when they go to the filling station. Just as a vehicle needs fuel in order to function, so too does a data-driven AI tool. Moreover, the concept of fuel can be used to foster discussions about both the quantity and the quality of data needed for the AI tool to perform well. For instance, a car's

performance could go down when the fuel tank is low (i.e., a low-resource situation), but it can go up when more fuel is added (i.e., a high-resource situation) (see 2.1.2). And in addition to needing enough fuel, a car will perform best when it has the right type and quality of fuel. Using the wrong type of fuel, or lower quality of fuel, will negatively affect the car's performance.

4.3 Starting with image data

4.3.1 Machine learning

The science communication tips also recommend incorporating visuals as a means of supporting explanations. In the case of AI tools, it might be helpful for educators to use image-based data as a way to explain machine learning before moving on to translation data.

Machine learning is an AI technique that is used by translation tools, chatbots, predictive text, image generators, and more (Flach2012). It is a subfield of AI that enables computers to learn things that have not been explicitly programmed. Machine learning begins with data. The type of data needed depends on the task that the AI tool is being asked to do. Machine learning has four main steps:

- Identify a task for the AI tool.
- Show the tool examples of what you want it to learn (i.e., the training corpus).
- Give the tool some feedback (e.g. confirm the correct answers).
- Test the tool on new data that it has not seen before.

The following sections contain some examples that can be shared with translation students to help them understand how AI tools interact with data as part of machine learning.

4.3.2 Example 1: The lone banana problem

In this example, the task for the AI tool is to generate images – a concept that is already likely to be familiar to students since some of the earliest best-known GenAI tools were for image generation (e.g. DALL-E, Midjourney). For the purpose of this example, the task can be narrowed down even more to generating images of bananas.

To see examples of images of bananas, translation educators and students can navigate to open access image sites such as Pixabay.com or Unsplash.com and to use the search term “bananas”. To perform well, an AI tool would need to be provided with many, many examples of pictures of bananas in order to be able to generate a new image based on these examples. People typically need fewer examples from which to learn.

A fun exercise to do in class is to ask the students to look at a dozen pictures of bananas on an open source site, and then to imagine that they are an AI image generation tool that has been asked to generate (i.e., draw) their own image in response to the following prompt: “Draw a banana.”

This will not likely pose too much of a challenge for the students – even those who are not artistically inclined! As a follow up, share with students the results of Daniel **Hook2023**’s (**Hook2023**) experiments in which he asked the GenAI tool Midjourney to generate images of a banana. What Hook found was that the AI tool only generated images with two or more bananas, rather than a single banana. Hook refined the prompt several times asking for “a banana”, “a single banana”, “one banana”, and so on, but the tool continued to generate images containing multiple bananas.

This example, which **Hook2023** dubs “The lone banana problem”, can be used as a simple yet effective way of explaining the issue of data bias. If the training data does not contain a sufficient number of images of individual bananas, then the AI tool will not be able to generate an image of a single banana. Of course, the problem can be fixed by adding more data (i.e., images of lone bananas), but as Hook warns, we do not always realise what gaps are there. **Hook2023** also stresses that it is important to recognise that AI tools do not understand the world the way that people do: they only understand commonly occurring patterns.

While humans are amazing pattern matchers, that skill is augmented by common sense (in many but not all cases), context and an evolved and subtle understanding of the physical world around us. AIs don’t yet have those augmentations – they are pure pattern matching power. And hence, they are only as good as the data that we input into the training set and hence can be no more than the statistical average of those inputs. In the lone banana problem, the statistics suggested that bananas only appear in twos (or more) and so the AI could not imagine a single banana. (**Hook2023**, n.p.)

4.3.3 Example 2: The snow detector

As a follow up to the lone banana problem, it may be useful to share the example of an AI tool whose task is image classification. This example comes from

RibeiroEtAl2016, who trained an AI tool using two different sets of images – ten images of husky dogs and ten images of wolves. After the training session, the researchers tested the AI tool’s ability to classify images correctly by showing it ten new images (five huskies and five wolves) that had not been part of the initial training data. The AI tool classified most of the images correctly, but it made two errors. In investigating the cause of the errors, the researchers found that the AI tool was making decisions not based on the animals in the pictures but based on the background. In the training data, all the pictures of the wolves contained snow, but none of the pictures of huskies contained snow. Therefore, in the test phase, if the image showed an animal against a snowy background, the AI tool classified the image as a wolf, but if there was no snow, then the AI tool classified the image as a husky. In other words, the AI tool was looking for patterns, but the pattern that it used as a basis for its decision-making was not a sensible one.

The snow detector is a funny example, but the medical literature contains more serious examples of similar phenomena. In **NarlaEtAl2018**, researchers tried to train an AI tool to distinguish between images of cancerous and non-cancerous skin lesions. However, in the training corpus, the majority of images of cancerous lesions also contained a ruler to measure the size of the lesion. During the testing phase, it became clear that the AI tool was equating the absence of a ruler with the absence of cancer. In another example, an AI system was trained to distinguish lung X-rays of people with pneumonia from lung X-rays of people with COVID-19 (**RobertsEtAl2021**). Unfortunately, the training data containing examples of pneumonia consisted mainly of pediatric patients, while the training data with examples of COVID-19 was from mainly adult patients. As a result, the AI tool learned to distinguish between children and adults rather than between pneumonia and COVID-19.

4.3.4 Connecting the dots: from images to translation

Though humorous, the lone banana and snow detector examples are intended to drive home the very serious point that AI tools do not think, understand or even process information the way that people do. By focusing on relatively straightforward, concrete and visual tasks such as image generation and image classification, these examples are easy for students to relate to, and they can then be connected to translation-related issues. For example, the lone banana problem can lead into a discussion of data bias and of what happens when a translation corpus has gaps in the training data. For instance, **VanmassenhoveEtAl2018** have demonstrated how data bias can lead to issues of gender bias in translation, while

BowkerBlain2022 point out how a lack of data in one language variety can lead to content from another variety being used erroneously in a translation.

In addition to reinforcing the lesson about data bias, the snow detector example, along with the more serious examples of the ruler detector and the child detector, can be used to promote discussions on opacity and what can go wrong when it is not clear what patterns the AI tool is using as the basis for decisions. These examples can also feed discussions about risk assessment – an important element of any decision about whether or not to use an AI translation tool (**KoponenNurminen2024**). In some cases, the translation task at-hand might be relatively low-stakes (e.g. translating a text for the purposes of entertainment), and so using an AI tool could be a good option. However, in other circumstances, the translation task might be one with higher stakes, where the consequences of a poor translation could negatively affect someone's health or life. Another type of information that can be important as part of risk assessment is understanding whether the situation is more likely to be a high-resource or a low-resource situation, given that the performance of the AI tool is likely to be better for high-resource languages, language varieties, language pairs and domains, and poorer in low-resource situations (see 2.1.2).

5 Conclusion

AI translation tools are rapidly inserting themselves into the translation industry and must also be addressed in translator education. The technology behind these tools is very sophisticated and may be intimidating for translation educators and students alike, given that most do not have a background in computer science. However, most language professionals do not need to understand all the details of how AI translation tools work in order to use them in an informed and responsible way. In contrast, it is extremely useful for language professionals to understand how these tools interact with data, and how the data used for machine learning can affect tool performance and translation quality. In the context of AI translation tools, the data is typically a corpus. Translation students are already familiar with corpora, which are also used with other types of translation tools (e.g. concordancers, term extractors, translation memory systems). However, students need to learn how AI tools are impacting the characteristics of corpora.

Rather than approaching the teaching of AI translation tools from the more traditional perspective of scientific or expert-to-expert communication, translation educators may find that science communication techniques (e.g. framing,

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analogies, visualisation) are more useful for teaching students about data and machine learning.

In the spirit of transversal learning, translation educators may even find that developing effective techniques to teach their translation students about data and its role in AI translation tools can pay off in other ways. As **Borowiec2023** laments,

Despite there being many excellent reasons for scientists to engage in science communication, they often lack the tools to do so. [...] Scientists frequently cite lack of training and/or confidence in their science communication skills as a barrier to their participation in public-facing activities.

Since many translator educators are also researchers, they may be able to apply these science communication techniques to their research-oriented activities. Moreover, **Ehrensberger-DowEtAl2023** suggest that translator trainers are well placed to take on a role such as machine translation literacy consultant, which could involve helping clients (who are unlikely to be computer scientists) to understand how AI tools process data or how data resources can be curated to improve the tools' performance.